

NRES 746 – Week 4, Lecture 1: Tracing the Binomial

Name: _____

Instructions: Work alone for about 5-8 minutes. Then compare with a neighbor.

Part 1: Trace the algorithm by hand

`dbinom()` is not magic – it’s just the binomial pmf,

$$P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k},$$

evaluated one piece at a time. Trace through it by hand for $N = 5$, $k = 2$, $p = 0.4$, filling in each step **before** moving to the next.

Step 1. Compute $\binom{N}{k} = \binom{5}{2}$ (the number of ways to choose which 2 of the 5 trials are “successes”):

$$\binom{5}{2} = \underline{\hspace{2cm}}$$

Step 2. Compute $p^k = 0.4^2$:

$$0.4^2 = \underline{\hspace{2cm}}$$

Step 3. Compute $(1 - p)^{N-k} = 0.6^3$:

$$0.6^3 = \underline{\hspace{2cm}}$$

Step 4. Multiply the three pieces from Steps 1-3 together:

$$P(X = 2) = \underline{\hspace{2cm}}$$

Part 2: Find the bug

A student wrote the following R function to compute the binomial pmf completely from scratch – no `dbinom()` allowed. It has exactly **one** bug. Circle it, and write the corrected line below.

```
binom_pmf <- function(N, k, p) {  
  choose(N, k) * p^k * (1 - p)^(N - k - 1)  
}
```

Corrected line:

2b. If you ran the buggy version with `binom_pmf(5, 2, 0.4)`, would it return a number bigger or smaller than your hand-traced answer from Part 1? How can you tell without running it?
